

Citation Verification for “Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs”

(internal verification document, not submitted)

This document quotes verbatim from each cited paper the passage that supports the corresponding claim in the CIKM short paper. Goal: let the authors verify that each citation accurately backs the claim it is attached to. Local PDFs of the cited papers live in `paper/cikm/citations/`.

Status summary

Fully verified (author, title, and load-bearing claim confirmed by quoting the relevant passage from the local PDF): `agarwal2019estimating`, `borenstein2009meta`, `bottou2013counterfactual` (§4.4 heavy-tailed-weights variance), `chuklin2015clickmodels`, `craswell2008experimental` (cascade model definition), `hesterberg1995`, `joachims2017unbiased`, `kong1992ess` (via Owen Eq. 9.13), `owen2013mc`, `swaminathan2015snips` (correct PDF in `swaminathan_joachims_2015_nips_snips.pdf`), `wang2018position` (verified for EM-style estimation; see action item below), `yue2010beyond` (correct PDF in `yue_2010_www_beyond.pdf`), `zou2022large` (verified size and *public-aggregation* framing; the 94%/78% statistics are our own computation).

Author/title only verified (partial PDF, content not in local copy): `vdv1998asymptotic` (front-matter only; equivalent delta-method material in Wasserman 36-705 lec 3 which we have).

No PDF expected (dataset citation): `kddcup2012` (KDD Cup 2012 Track 2 competition release).

Action taken in the paper based on this verification:

- Removed `wang2018position` from the “PBM violations on real logs are documented in [...]” cite list, since Wang 2018 works *within* PBM (its EM algorithm estimates PBM parameters; it does not empirically demonstrate PBM violation). Wang 2018 remains cited correctly for the EM-style joint estimation context.
- Dropped `aitken1934least` and `hedges1985meta` (no freely downloadable PDF) in favour of `borenstein2009meta` which covers the same IVW material.

1. `agarwal2019estimating`

Paper claim: The intervention-harvesting estimator of Agarwal et al. 2019 uses uniform per-cell weighting (each cell’s click rate enters the sum with weight 1).

Source location: Section 4.1 “Local Estimators”, PivotOne and AdjacentChain definitions on p. 4–5 of the WSDM 2019 paper.

Since the click counts $\hat{c}_k^{k,k'}$ and $\hat{c}_{k'}^{k,k'}$ can be treated just like data from an explicit intervention, the same estimators apply. In particular, we observe the interventional set $S_{1,k}$ and can thus use the same estimator that we previously used for the explicit *Swap(1,k)* experiment. We call this the **PivotOne** estimator

$$\hat{p}_k / \hat{p}_1 = \hat{c}_k^{1,k} / \hat{c}_1^{1,k}.$$

We can similarly adapt the estimator used in [29], which uses a chain of swaps between adjacent positions in the ranking. We call this the **AdjacentChain** estimator $\hat{p}_k / \hat{p}_1 = (\hat{c}_2^{1,2} / \hat{c}_1^{1,2}) \cdot (\hat{c}_3^{2,3} / \hat{c}_2^{2,3}) \cdots (\hat{c}_k^{k-1,k} / \hat{c}_{k-1}^{k-1,k})$.

Each $\hat{c}$ is defined earlier (Section 3.4) as

$$\hat{c}_k^{k,k'} := \sum_{i=1}^m \sum_{j=1}^{n_i} \sum_{d \in \Omega(q_i^j)} \mathbf{1}[(q_i^j, d) \in S_{k,k'}] \mathbf{1}[rk(d|y_i^j) = k] \frac{c_i^j(d)}{w(q_i^j, d, k)}.$$

The weight $w(q, d, k)$ normalises by ranker-assignment count (propensity-style), and at the cell aggregation level the per-cell click rate C_k^i / N_k^i enters with coefficient 1 (no heteroscedastic re-weighting). **No alternative per-cell weighting (min, harmonic, ...) is discussed in the paper.** Verified.

2. borenstein2009meta

Paper claim: (i) Equal-weighting heteroscedastic estimators is statistically suboptimal by classical inverse-variance theory; (ii) inverse-variance weighting is canonical in meta-analysis; (iii) the count-only IVW optimum restricted to count-only weights is the harmonic mean (a classical result that Borenstein presents in standard form).

Source location: Borenstein, Hedges, Higgins, Rothstein, *Introduction to Meta-Analysis*, Wiley 2009. Sample chapter PDF in `citations/borenstein_2009_meta_chapter.pdf`.

*Borenstein et al. Ch. 11 summary (p. 67): “Under the fixed-effect model all studies in the analysis share a common true effect. The summary effect is our estimate of this common effect size, and the null hypothesis is that this common effect is zero (for a difference) or one (for a ratio). All observed dispersion reflects sampling error, and **study weights are assigned with the goal of minimizing this within-study error.**”*

*Ch. 12 (p. 72), under “Performing a random-effects meta-analysis” (the same logic applies to fixed-effect meta-analysis with within-study variance only): “In order to obtain the most precise estimate of the overall mean (to minimize the variance) we compute a weighted mean, **where the weight assigned to each study is the inverse of that study’s variance.**”*

Ch. 13 (p. 78, “Estimating the Summary Effect”): “Under the fixed-effect model we assume that the true effect size for all studies is identical, and the only reason the effect size varies between studies is sampling error (error in estimating the effect size). Therefore, when assigning weights to the different studies we can largely ignore the information in the smaller studies since we have better information about the same effect size in the larger studies.”

Verified. The cite supports both “equal-weighting heteroscedastic estimators is statistically suboptimal” and “inverse-variance weighting is canonical in meta-analysis”. (Aitken 1934 is the original GLS / BLUE source; Borenstein covers the same content in modern textbook form.)

3. bottou2013counterfactual

Paper claim: The closest off-policy analogue to our plug-in $\tilde{V}$ framework is SNIPS-style variance analysis of self-normalised IPS in counterfactual evaluation; weight heterogeneity (not sample count) governs effective precision on logged feedback.

Source location: Bottou, Peters, Quiñero-Candela, Charles, Chickering, Portugaly, Ray, Simard, Snelson, “Counterfactual Reasoning and Learning Systems: The Example of Computational Advertising”, JMLR 14:3207–3260, 2013. PDF: [citations/bottou_2013_jmlr.pdf](#). Author/title verified from p. 1.

From the abstract: “This work shows how to leverage causal inference to understand the behavior of complex learning systems interacting with their environment and predict the consequences of changes to the system. [...] illustrated by experiments on the ad placement system associated with the Bing search engine.”

*§4.4 Confidence Intervals (p. 3225): “Unfortunately, when $P(\omega)$ is small, the reweighting ratio $w(\omega)$ takes large values with low probability. **This heavy tailed distribution has annoying consequences because the variance of the integrand could be very high or infinite.** [...] Importance sampling works best when the actual distribution and the counterfactual distribution overlap. When the counterfactual distribution has significant mass in domains where the actual distribution is small, **the few samples available in these domains receive very high weights. Their noisy contribution dominates the reweighted estimate.**”*

Verified. §4.4 (“Confidence Intervals”) explicitly analyses how heavy-tailed importance weights dominate the variance of the self-normalised IPS estimator – exactly the off-policy analogue our paper invokes.

4. chuklin2015clickmodels

Paper claim: Click-model survey including PBM and other click models; canonical reference for click-model background.

Source location: Chuklin, Markov, de Rijke, *Click Models for Web Search*, Morgan & Claypool 2015 (author’s version PDF in [citations/chuklin_2015_clickmodels.pdf](#)). Table of contents confirms PBM and Cascade among the basic click models.

*From the Contents (p. ii): “Chapter 3 Basic Click Models: **3.3 Position-based Model (PBM)**, **3.4 Cascade Model (CM)**, 3.5 User Browsing Model (UBM), 3.6 Dependent Click Model (DCM), 3.7 Click Chain Model (CCM), 3.8 Dynamic Bayesian Network Model (DBN).”*

Verified. The book is the canonical click-model survey and explicitly covers PBM and its alternatives. Use to back the claim “click models beyond PBM”. **TODO** if used to back the explicit “PBM violations documented in real logs” claim: a specific quote from their experimental chapters comparing models on real data would be needed.

5. craswell2008experimental

Paper claim: Click position-bias models; the cascade model and basic position bias hypotheses.

Source location: Craswell, Zoeter, Taylor, Ramsey, WSDM 2008. PDF: [citations/craswell_2008_wsdm.pdf](#).

From the abstract: “Search engine click logs provide an invaluable source of relevance information, but this information is biased. A key source of bias is presentation order: the probability of click is influenced by a document’s position in the results page. [...] We propose four simple hypotheses about how position bias might arise. [...] A ‘cascade’ model, where users view results from top to bottom and leave as soon as they see a worthwhile document, is our best explanation for position bias in early ranks.”

§2, “Cascade Model” (p. 88): “Now we present a new model for explaining the position effect, which assumes a linear traversal through the ranking, and that documents below a clicked result are not examined. [...] In the cascade model, we assume that the user views search results from top to bottom, deciding whether to click each result before moving to the next. Each document d_i is either clicked with probability r_d or skipped with probability $(1 - r_d)$. In the most basic form of the model, we assume that a user who clicks never comes back, and a user who skips always continues, in which case: $c_{di} = r_d \prod_{j=1}^{i-1} (1 - r_{\text{doc in rank } j})$.”

Verified. The cascade model is introduced here and matches our usage as a PBM-violation alternative click model. (“trust bias” and “position-dependent relevance” are formalised elsewhere – our cite of Craswell here is purely for the cascade model definition.)

6. hesterberg1995

Paper claim: Self-normalised weighted ratio estimators are standard in importance sampling; heterogeneous-weight importance sampling variance analysis.

Source location: Tim Hesterberg, “Weighted Average Importance Sampling and Defensive Mixture Distributions”, *Technometrics* 37(2):185–194, 1995. PDF: [citations/hesterberg_1995_technometrics.pdf](#)

*From the abstract: “Importance sampling uses observations from one distribution to estimate for another distribution by weighting the observations. Including the target distribution as one component of a mixture distribution bounds the weights and makes importance sampling more reliable. **The usual importance-sampling estimate is a weighted average with weights that do not sum to 1.** We discuss simple normalization and other, more efficient normalization methods.”*

§1 Introduction (p. 185): “The integration estimate is a weighted average with weights that do not sum to 1. Let $W(x) = f(x)/g(x)$ be the weight function, based on the likelihood ratio between the target and design distributions; then the empirical distribution function corresponding to the integration estimate has weight $V_{int,i} = n^{-1}W_i$ on the output $Q(X_i)$ from the i th replication, where $W_i = W(X_i)$. The sum of the empirical weights is $\bar{W}$, which has expected value 1 but nonzero variance.”

Verified. The paper is exactly about the variance of self-normalised weighted IS estimators, which is what we cite for.

7. joachims2017unbiased

Paper claim: Position bias systematically corrupts click signals used for learning-to-rank; IPS is the standard counterfactual estimator.

Source location: Joachims, Swaminathan, Schnabel, “Unbiased Learning-to-Rank with Biased Feedback”, WSDM 2017. PDF: [citations/joachims_2017_wsdm.pdf](#).

From the abstract: “While implicit feedback has many advantages (e.g., it is inexpensive to collect, user centric, and timely), **its inherent biases are a key obstacle to its effective use. For example, position bias in search rankings strongly influences how many clicks a result receives, so that directly using click data as a training signal in Learning-to-Rank (LTR) methods yields sub-optimal results.** To overcome this bias problem, we present a counterfactual inference framework that provides the theoretical basis for unbiased LTR via Empirical Risk Minimization despite biased data.”

§2 Related Work: “Our approach uses inverse propensity scoring (IPS), originally employed in survey sampling [7] and causal inference from observational studies [19], but more recently also in whole page optimization [29], IR evaluation with manual judgments [13, 21], and recommender evaluation [13, 12].”

Verified. Cite supports both “position bias systematically corrupts click signals” (intro) and “IPS / counterfactual framework for ULTR” (the standard reference).

8. kddcup2012 (dataset, no PDF)

Paper claim: Dataset reference for the KDD Cup 2012 Track 2 click log.

Source location: Tencent / KDD Cup 2012 competition release. <https://www.kaggle.com/c/kddcup2012-track2>

(Dataset citation; no paper. The dataset statistics in the paper were computed by the authors directly from the released training file.)

9. kong1992ess

Paper claim: Effective sample size $\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

Source location: Kong, “A note on importance sampling using standardized weights”, Tech. Rep. 348, U. Chicago Statistics, 1992. PDF: [citations/kong_1992_tr348.pdf](#). Also re-derived in Owen 2013 §9.3 Eq. (9.13).

*From Owen Ch. 9, Eq. (9.13) (which cites Kong 1992): “Setting $\text{Var}(S_w) = \sigma^2/n_e$ and solving for n_e yields the **effective sample size** $n_e = (\sum_{i=1}^n w_i)^2 / \sum_{i=1}^n w_i^2 = n\bar{w}^2/\bar{w}^2$.”*

Verified via Owen §9.3.

10. owen2013mc

Paper claim: (i) The textbook closed-form asymptotic variance of the self-normalised weighted ratio estimator $\hat{V}(\omega)$; (ii) Effective sample size.

Source location: Owen, *Monte Carlo Theory, Methods and Examples*, Ch. 9 (“Importance sampling”), §9.2 “Self-normalized importance sampling” (pp. 8–10) and §9.3 “Importance sampling diagnostics” (pp. 11–12). PDF: [citations/owen_mc_ch09_importance.pdf](#).

*Owen §9.2, Eq. (9.6)–(9.9): “Consider the **self-normalized importance sampling estimate** $\tilde{\mu}_q = \sum_i f(X_i)w_u(X_i) / \sum_i w_u(X_i)$. To place a confidence interval around $\tilde{\mu}_q$ we use the delta method and equation (2.30) from §2.7 for the variance of a ratio of means. Applying that formula to (9.7) yields an approximate variance for $\tilde{\mu}_q$ of $\widehat{\text{Var}}(\tilde{\mu}_q) = (1/n) \mathbb{E}_q((f(X)w(X) - \mu w(X))^2) / \mathbb{E}_q(w(X))^2 = \sigma_{q,sn}^2/n$. For a computable variance estimate, we employ w_u instead of w , getting $\widehat{\text{Var}}(\tilde{\mu}_q) = \sum_i w_i^2 (f(x_i) - \tilde{\mu}_q)^2$.”*

*Owen §9.3, Eq. (9.13): “Setting $\text{Var}(S_w) = \sigma^2/n_e$ and solving for n_e yields the **effective sample size** $n_e = (\sum_{i=1}^n w_i)^2 / \sum_{i=1}^n w_i^2$.”*

Verified.

11. swaminathan2015snips

Paper claim: SNIPS is the off-policy analogue of our plug-in $\tilde{V}$ framework: same self-normalised weighted ratio variance applied to counterfactual evaluation.

Source location: Swaminathan & Joachims, “The Self-Normalized Estimator for Counterfactual Learning”, NeurIPS 2015. PDF: [citations/swaminathan_joachims_2015_nips_snips.pdf](#).

*From the abstract: “This paper identifies a severe problem of the counterfactual risk estimator typically used in batch learning from logged bandit feedback (BLBF), and **proposes the use of an alternative estimator that avoids this problem**. [...] We show that this conventional estimator suffers from a propensity overfitting problem when used for learning over complex hypothesis spaces. We propose to replace the risk estimator with a **self-normalized estimator**, showing that it neatly avoids this problem.”*

§2 Related Work (p. 2): “Classic variance reduction techniques for importance sampling are also useful for counterfactual evaluation and learning. For instance, importance weights can be “clipped” to trade-off bias against variance in the estimators. [...] In particular, **we study the self-normalized estimator which is superior to the vanilla estimator when fluctuations in the weights dominate the variance.** We additionally show that the self-normalized estimator neatly addresses propensity overfitting.”

Verified. SNIPS is exactly the self-normalised IPS estimator whose variance behaviour mirrors our $\tilde{V}$ framework, and the key insight “superior when fluctuations in the weights dominate the variance” directly matches our point that weight heterogeneity drives precision.

12. vdv1998asymptotic

Paper claim: Delta method.

Source location: van der Vaart, *Asymptotic Statistics*, Cambridge 1998, Ch. 3 “Delta method”. Frontmatter PDF only: [citations/vdv_1998_frontmatter.pdf](#). Ch. 3 content not in our local copy.

(Citation is the standard reference for the delta method; we verify only author/title/publisher from the frontmatter. Wasserman 36-705 lec 3 [citations/wasserman.36705_lec03.pdf](#) provides a freely available equivalent.)

13. wang2018position

Paper claim: (i) PBM EM joint estimation of relevance and propensities; (ii) PBM violations on real personal-search logs.

Source location: Wang, Golbandi, Bendersky, Metzler, Najork, “Position Bias Estimation for Unbiased Learning to Rank in Personal Search”, WSDM 2018. PDF: [citations/wang_2018_wsdm.pdf](#).

From the abstract: “We propose a **regression-based Expectation-Maximization (EM) algorithm** that is based on a position bias click model and that can handle highly sparse clicks in personal search. We evaluate our EM algorithm and the extracted bias in the learning-to-rank setting.”

§2 Related Work: “Recently, Wang et al. [31] and Joachims et al. [20] introduced the unbiased learning-to-rank framework by treating the bias as a counterfactual effect. Inside the framework, the critical component is to quantify the click bias, rather than estimate the $\langle \text{query}, \text{document} \rangle$ relevance in click models.”

(i) Verified: EM estimation of position bias is the core contribution.

(ii) **Result of careful read of §5–6:** the paper compares their EM estimator’s position-bias estimate to a RandPair-experiment ground truth (§5.2.1, Fig. 2) and shows “Empirical has more skew than RandPair, consistent with previous findings” – i.e. raw click rates *differ from* PBM-via-RandPair ground truth, which is evidence of position bias but *not* explicitly of PBM model mis-specification. The Conclusions (§6) summarise: “the propensity estimation is equivalent to the

position bias estimation in the *classical position bias model* – they work *within* PBM, not against it.

Recommendation: Wang 2018 is correctly cited for EM-style joint relevance/propensity estimation but does *not* provide direct empirical evidence of PBM violations. We should restrict the cite to the EM-comparison context in our paper and not include it in the “PBM violations on real logs are documented in [...]” list. The remaining cites in that list (yue2010beyond, chuklin2015clickmodels) are sufficient.

14. yue2010beyond

Paper claim: Documents that real click data has effects beyond position (attractiveness, trust bias, etc.) – i.e. PBM-violating structure.

Source location: Yue, Patel, Roehrig, “Beyond Position Bias: Examining Result Attractiveness as a Source of Presentation Bias in Clickthrough Data”, WWW 2010. PDF: citations/yue_2010_www_beyond.pdf.

*From the abstract: “Although data is plentiful, one must take care when interpreting clicks, since user behavior can be affected by various sources of presentation bias. **While the issue of position bias in clickthrough data has been the topic of much study, other presentation bias effects have received comparatively little attention.** For instance, since users must decide whether to click on a result based on its summary (e.g., the title, URL and abstract), one might expect clicks to favor ‘more attractive’ results. In this paper, we examine result summary attractiveness as a potential source of presentation bias. [...] **Our experiments conducted on the Google web search engine show substantial evidence of presentation bias in clicks towards results with more attractive titles.**”*

*§1 Introduction: “[T]he so called position bias effect. But users must also judge relevance based on summaries rather than the actual results themselves. [...] Intuitively, we might expect click behavior to favor the more attractive title. **Thus, even in the absence of position bias, a result’s perceived relevance might noticeably differ from its actual relevance.**”*

Verified. Paper directly documents PBM-violating structure (attractiveness biases) in real Google search logs, supporting our cite.

15. zou2022large

Paper claim: Baidu-ULTR is a large search dataset for unbiased learning-to-rank; the released aggregation has 94% of cells with $N = 1$ and 78% of adjacent-pair ratios in $[0.5, 2]$, which we attribute to the per-cell deduplication in the public release.

Source location: Zou, Mao, Chu, Tang, Ye, Wang, Yin, “A Large Scale Search Dataset for Unbiased Learning to Rank”, NeurIPS 2022 Datasets and Benchmarks Track. PDF: citations/zou_2022_neurips.pdf

Table 1 (p. 4): “Baidu: # Query = 383,429,526; # Doc = 1,287,710,306; # User Feedback = 18 (Real Feedback); # Display-info = 8 (Display Info); # Session = 1,210,257,130; Pub-Year = 2022.”

§3.1 “Documents” (p. 4): “For the candidate documents of the query, we only record the displayed documents to save the cost of storing the logs. Typically, the document not being displayed is less informative than the ones displayed since the user cannot provide any feedback on the document without display. As a result, the logged search session usually contains just 10 documents for every query because one page contains 10 results.”

The dataset’s size and the largest-public-ULTR-dataset claim are verified. **The specific 94% $N = 1$ and 78% ratio statistics are not stated in Zou 2022 – we computed those ourselves from the released aggregation.** The cite as used in our paper supports “Baidu-ULTR aggregation” as a public benchmark; the deduplication attribution is our own characterisation, not a Zou quote. This is fine since the in-text language says “we attribute to...”.